CSI4901- Capstone Project

MULTIMODAL FAKE NEWS DETECTION USING CNN AND TRANSFORMER MODELS

Submitted in partial fulfilment of the requirements for the degree of

Master of Technology [Integrated]

in

Computer Science and Engineering (Data Science)

by

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(Deemed to be University under section 3 of UGC Act, 1956)

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DECLARATION

I hereby declare that the project entitled "MULTIMODAL FAKE NEWS DETECTION USING CNN AND TRANSFORMER MODELS" submitted by me, for the award of the degree of Master of Technology [Integrated] in Computer Science and Engineering (Data Science) to VIT is a record of bonafide work carried out by me under the supervision of Prof. Vijayasherly V, School of Computer Science and Engineering.

I further declare that the work reported in this project report has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Vellore **SRIDEV S (21MID0051)**

Date :

CERTIFICATE

This is to certify that the project entitled "MULTIMODAL FAKE NEWS DETECTION USING CNN AND TRANSFORMER MODELS" submitted by SRIDEV S (21MID0051), School of Computer Science and Engineering, VIT, for the award of the degree of Master of Technology [Integrated] in Computer Science and Engineering (Data Science), is a record of bonafide work carried out by the student under my supervision during academic year 2025-2026, as per the VIT code of academic and research ethics.

The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university. The project fulfills the requirements and regulations of the University and in my opinion meets the necessary standards for submission.

Place: Vellore **Signature of the Guide**

Date :

Internal Examiner **External Examiner**

Head of the Department

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EXECUTIVE SUMMARY

The quick progress in digital media and the development of web platforms have made it possible for news to reach many people without compromising accessibility, but the flow of false and fake news has increased. This has resulted in a complex problem to identify false news given the new format of media contents which blend visual and textual information. The following shows a system that has been developed for the detection of Multimodal Fake News.

The suggested system follows a hybrid methodology in which deep learning techniques along with AI based validation are incorporated. A transformer based model is employed to perceive contextual and semantic content from text data, while an additional convolutional neural network extracts useful visual information from images. Both these features are then combined using a multimodal fusion technique, for generating the initial prediction. For increased robustness, a separate AI validation layer is introduced which conducts semantic reasoning and credibility verification, to generate a secondary prediction. Finally, a decision mechanism combines both these predictions, to generate the final output.

The model was trained and evaluated on a balanced dataset containing 9000 samples of news and produced great results in terms of accuracy in predicting fake news. Additionally, the system has been built into an interactive and easy to use web application which can be used to verify news in real-time. With features such as a confidence meter and support for multiple languages, user interaction is made easier and it also allows the users to understand more about the confidence level in a prediction.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
NLP	Natural Language Processing
CNN	Convolutional Neural Network
MLP	Multi-Layer Perceptron
DFD	Data Flow Diagram
UML	Unified Modeling Language
API	Application Programming Interface
URL	Uniform Resource Locator
UI	User Interface
BERT	Bidirectional Encoder Representations from Transformers
DistilBERT	Distilled version of BERT
ResNet	Residual Network
XAI	Explainable Artificial Intelligence

CHAPTER 1
INTRODUCTION

1.1 BACKGROUND

Digital communication technology develops fast, it's now widely spread over almost every field. The massive usage of social network has completely changed the world concerning information production, spreading, and consummation. Now, a single piece of information can travel up to million users in few seconds, which is faster and easier for everyone. But one huge problem has been brought: rapid and easy spreading of fake and harmful information. Fake news is a specific term for purposely false or deceptive content stated as legitimate news, the consequence could impact public belief, political safety and the overall condition of society in several ways [1],[18],[40]. There are too many information generated by internet users in an extremely short time for us to check manually, this make manual methods unable to work in such cases [18],[24].

Most traditional methods to detect fake news use NLP techniques. NLP techniques analyze the content of the text. The advantages of these methods are that they recognize patterns, tones and the context of words used. These techniques ignore visual information, such as images, which help in drawing the user's attention and increase the fake message spreading [3], [27]. CNNs-based image detection algorithms help in detecting inconsistencies in an image or visual modifications, but ignore the meaning of news [11], [25].

Fake news does not consist of fake texts only. It is very usual that fake news consists of fake text and false images; therefore the analysis of multiple pieces of information in an integrated fashion is also needed. Based on the facts that real fake news is almost always multimodal, a multimodal system with analysis of texts and images together is needed to build a more robust fake news detection system [2], [6], [24]. Recently, advanced deep learning techniques have led to effective feature representation for text and image respectively; especially transformer-based language models (BERT, etc. [20]) andCNN-based image analyses [12]. The integration of both these deep learning models can create a very strong system.

1.1.1 Overview of Fake News and Misinformation

Fake news denotes the deliberate propagation of false, misleading, or deliberately fabricated information across online platforms including social media, online news sites, blogs and mobile applications. Unlike misinformation, which could be unintentional, fake news is often constructed with a certain intention behind the information to affect people's perspective, and thus create social or political discrimination or to lure people into seeing an alarming story. The spread of deceptive content has become of immense importance in the present digital world, in which information is transmitted fast and rapidly spread over platforms.

The rapid development of the internet and social media has changed completely how news is produced and used. On one hand, news sources have become easier to reach and communicate with. But on the other, the Internet and the social media allow false news to circulate much faster and on a bigger scale than ever before. Within seconds, the news can reach millions, sometimes even without being verified. This ease of spreading the news as well as users' inclination to repost sensational and emotive content contribute to the speed and effects of false news.

The significant difficulty in dealing with fake news is how the stories may affect the perceptions and decision-making abilities of the public. They can alter public opinions regarding various social and health issues, public policies, politics, etc, causing public misapprehension or even significant harm to people's lives. During times of worldwide events or global crises, a stream of fake news might produce panic and false claims, shaking public faith in authoritative sources.

In addition, the increased development of digital content creation tools have also made it easier to produce highly convincing fake content such as edited images, fake headlines and made up stories. Fake content, in fact, is usually very persuasive, using both fake and convincing language as well as attractive images, making it difficult for humans to detect. Human methods for verification are therefore no longer enough for the volume and complexity of fake news we encounter today.

However, with these growing difficulties, there is an emerging demand for automatically intelligent systems capable of reliably and rapidly identifying false news. These systems would have to not only rely on text information but also on the visual content and context as well in order to look for any incoherence and deceiving traits. The creation of such a reliable fake news detector system is then crucial to protect information authenticity and public credibility.

1.1.2 Existing Fake News Detection Approaches

Many various techniques have been proposed during years to overcome the challenge of fake news detection, ranging from traditional machine learning techniques to the recent deep learning techniques. The earlier models were mainly based on text-based analysis using features like frequency of words, sentences and sentence structure, along with other linguistic features, and employing algorithms from traditional machine learning like Nave Bayes, Support Vector Machine (SVM), and Logistic Regression. Though these techniques are useful to certain extent, they may not effectively handle the underlying context.

As deep learning technology keeps progressing, more advanced models are developed to boost the detection performance. Specifically, the transformer-based models such as DistilBERT have made remarkable progress in understanding contextual and semantic information in text. Different from traditional methods which analyze text only in isolation, the transformer-based models are aware of the surrounding text a word or sentence exists in. Thus they are better able to identify linguistic cues, deceptive language and implicit intention behind news headlines and articles. With those models, the performance of text-based fake news detection systems have significantly improved.

Apart from text analysis, other image based detection methods have also been studied for detection of deceptive/faked images. Methods such as Convolutional Neural Networks(CNNs), for instance,ResNet-18, are commonly used for visual feature extraction of an image and are adept at finding abnormal patterns or anomalies, especially for images used in fake news. On the downside, image-only models do not capture a contextual understanding of how the images relate to the text

More recently, multimodal methods, which utilize both text and visual content for detection, have been investigated. By learning multi-modal features, these systems have the ability to extract more comprehensive representations of the news contents and can possibly detect discrepancies. Although many previous systems have shown improvement compared to their single-modality counterparts, multimodal systems also suffer from computational overhead, difficulties of effectively combining the multi-modal features and interpretability.

All the above approaches demonstrate that considerable progress has been achieved in fake news detection, however, most of them only works on a single modality or lack of powerful reasoning capability. Advanced systems are in great need that can effectively utilize different data sources and give satisfactory, explicable results in real-world applications.

1.1.3 Limitations of Existing Systems

Although impressive progress has been made regarding fake news detection methods, there are several limitations of existing approaches that diminish their usefulness in real applications. A major weakness is the unimodal nature of most methods, with systems focusing on text only or visual information only. While text-based methods successfully discern linguistic cues that identify deceit, they do not discover contradictions that may be brought forth by a manipulated or incongruous image paired with text.Likewise image-based methods do not capture semantics to interpret what the associated context may be trying to convey.

Secondly, most traditional and some deep learning models lack significant contextual and reasoning power. Sophisticated architectures can extract features from data but these black-boxes perform classification without clear explanations. When classification is made using the system, the user finds it hard to grasp how it arrived at a specific decision. Hence, this results in low confidence about the system. It is paramount to have an intuitive explanation of the classification for real world applications such as news verification.

Besides, numerous current methods lack good generalization and flexibility. Fake news is a dynamically changing topic with new types, forms, and techniques continuously developing over time. Therefore, methods trained by static datasets usually achieve acceptable accuracy under controllable conditions but may fail when encountering unknown or new forms of misinformation. It shows the necessity of systems with adaptability.

Scalability and real-time processing-Social media, blogs, and other digital platforms are generating ever-increasing quantities of content daily. To keep pace with the speed and volume of information spread on these networks fake news detection systems must effectively and efficiently manage big data in real time. But, most existing fake news detection techniques are computationally complex, not adapted to run efficiently in real-time in production environments.

In addition, the lack of any kind of cross-validation mechanisms in many systems leads to potentially unreliable predictions. Given that most systems only learn from the existing data, there is nothing stopping a false positive from being predicted, and no further checks will occur, resulting in an unreliable prediction. A similar argument can be made for false negatives.

In summary, the deficiencies noted show that further work must be conducted to make the system more scalable, interpretable and robust. Most importantly, methods for data multimodal fusion and knowledge-based validation need to be introduced to detect new, emerging patterns.

1.1.4 Role of Multimodal Learning in Fake News Detection

As an effective technique to cope with drawbacks of conventional fake news detection system, multimodal learning utilizes information from more than one source data (i.e. Text, image, etc.) to analyze and investigate the news. In reality, no news can be represented by just one modality; news is often a combination of texts, image, videos and meta data for transmission of information. Hence, by analyzing only one modality might derive insufficient and/or incorrect deduction. However, in multimodal learning, the model can produce more rich and full representation by analyzing both the image and the texts.

Multimodal learning can also exploit inconsistencies in false data (cross-modal anomalies). For example, many false articles are presented with exaggerated or false headlines combined with unrelated/falsified images, aiming to generate public interest and establish greater confidence. A text-only approach is not guaranteed to notice this disjunction in text and image modalities; a multimodal approach analyzing both concurrently could potentially reveal such contradictions. This improves an information detection system's performance.

In multimodal scenarios, advanced model structures are used for respective modalities; transformer-based approaches are employed for textual data, while convolutional neural networks are generally used for images. These feature extracts are merged in a way that exploits information across modality using fusion mechanisms to provide a comprehensive feature representation encompassing both semantic and visual features. Enhanced performance in the classification tasks results from using information from both modalities, while overcoming their respective limitations.

Besides that, multimodal learning enhances the robustness and generalization capability of fake news detection systems. By integrating information from diverse modalities, the system becomes less susceptible to the limitations of any individual modality and hence more robust against noise, missing information or adversarial attacks. For instance, ambiguity or insufficiency in the textual modality can be compensated by the context extracted from visual information, and vice versa.

Another compelling advantage is the potential for enhanced interpretability. The varying levels of importance attributed to each modality in the classification process of a multimodal system can reveal useful insights as to why an article is flagged as fake or real. This is indeed becoming increasingly crucial with the need for explainable AI models in applications where it has a strong impact on decision making as well as in public opinion.

In conclusion, multimodal learning makes the task offake news detectioneasier by allowing more accurate, reliable and contextual analyses, and offers a path towards building effective solutions to the complex and widespread issue of manipulating digital information.

1.1.5 Need for an AI-Driven Multimodal System

As misinformation become increasingly sophisticated and extensive, there is a greater need for more intelligent, robust, and accurate fake news detection system. However, current methods have shown impressive performance but suffer from some drawbacks, such as their inability to exploit multimodal information and lack of explainability, to handle real-world fake news, which are becoming dynamic. Therefore, it is of interest to study systems that not only exploits data from multiple modes but also integrate a mechanism of validation that is based on reasoning. The contemporary fake news typically exploits many data types (texts, images, etc.) in order to appear credible, thus making it harder for systems dealing with only one modality to notice inconsistencies. Although powerful, state-of-the-art deep learning methods mainly detect fake news based on learned patterns and may not provide accurate predictions in the absence of logical reasoning and/or world knowledge, thus suggesting that it is necessary to create systems that combine data-driven learning with an intelligent reasoning component for validation.

The aforementioned drawbacks are countered by an AI-powered multimodal system that combines text and vision processing, as well as a level of higher-level reasoning. The text and vision processing rely on machine learning models like DistilBERT to process the text and ResNet-18 to process the image respectively, effectively extracting semantic as well as visual information. The crucial improvement here, however, is the addition of an AI-based validation layer which can perform text based functions such as, among others, semantic reasoning, assessment of language patterns, and assessing credibility of the source.

This validation layer serves as a second source of validation for ML model predictions to confirm their reliability by checking the accuracy of ML model predictions with another set of ML models and can enhance reliability by explaining decisions with the additional explanation of the decision. The interpretability that comes with a system like this enhances usability of the application because most systems lack the necessary explanations required to trust the machine learning models and also allows for users to justify decisions made from these decisions.

Moreover, real-time, scalable, user-friendly solutions are increasingly necessary. As the number of digital content continually increases, Fake news detection systems should be able to process large number of data efficiently and provide real-time feedback to the users. Furthermore, incorporating the system into an interactive interface such as a web application improves user-friendliness.

To conclude, an AI-based multimodal fake news detection system should be developed to resolve the aforementioned limitations. The combination of multimodal deep learning and reasoning based verification in an AI-based system could lead to more accurate and more robust, and at the same time, more interpretablefake news detection systems for real-world application.

1.2 MOTIVATION

The reason for working on this problem is due to the growing popularity of misinformation and the effects of the information on our society in the 21st century. The spreading offake newsis fast and reaches the audience as soon as it is disseminated on social networks due to the large spread and usage of these platforms. In a way,fake newsweakens our belief on true facts, causes unnecessary panics, influences public opinion, etc. [1], [40]. Thefake newsoften takes an appealing form consisting of pictures that we easily consider reliable [3], [27].

In previous fake news detection systems, mostly one type of data (unimodal approaches), such as text or image, has been used, and though these systems are useful in some specific situations, they are limited to consider the impact of text and image together as generally observed in real fake news [11] [25]. Many current research models are not very usable in real-time applications because they focus on accuracy but not on interpretability. It is also more desirable to have a system that can explain its prediction of fake news to make users trust it.

We want to develop an intelligent and scalable system, that is more human-friendly, thus trying to overcome these shortcomings of previous solutions. By the use of deep learning and multimodal approaches combined with an AI reasoning layer we are trying to increase the accuracy and robustness of the system. An AI validation layer based on a Large Language Model has been developed in order to increase the quality of decision-making by means of a reasoning layer with semantic knowledge and fact validation. To enhance applicability a language support and visualization component are available.

1.3 SCOPE OF THE PROJECT

This project covers the design, development and implementation of an end-to-end Multimodal Fake News Detection system where both the text and the images can be used as input. It will model the headlines of the news articles by a transformer model so as to achieve the semantic information and contextual meaning. Visual features from images can be represented by a CNN. The features from two modalities will then be merged by a late fusion approach, which feeds into a classification module to produce probability of prediction.

The described multimodal model is complemented with an AI based validation layer. It is used for the task of semantic reasoning, linguistic checking and credibility evaluation of sources. This layer provides an extra step of verification for increasing the accuracy of the predictions. The model uses a three stage decision pipeline in which the machine learning predictions and the AI based analysis are combined.

Additionally, we developed a web-based application with a Streamlit interface for interactive, real-time analysis. The application supports multiple languages with automatically detecting the input headlines and has a graphical confidence meter for users to intuitively gauge the accuracy of the predictions. The usage of the system is also being monitored using session analysis and performance is analyzed with a data of 9000 news samples and accuracy, precision, recall and F1-score as evaluation parameters.

Overall, it is believed that the described system is practical for real world applications of information verification, media credibility and digital content analysis, and is a user-friendly and scaleable approach for use within research.

CHAPTER 2

PROJECT DESCRIPTION AND GOALS

2.1 LITERATURE REVIEW

"Detecting Fake News and Disinformation Using Artificial Intelligence and Machine Learning to Avoid Supply Chain Disruptions" – Akhtar et al., 2023 [1]. This research uses supervised machine learning for news text analysis to identify misinformation in supply chain. It underscores the practical significance of misinformation in the real world, beyond social media. But the model only uses text and lacks visual and multi-modal features, which limits its ability to deal with current fake information.

"Multimodal Fake News Detection Using CNN and Transformer Models" – 2023 [2]. The researchers developed a multimodal approach that combines Convolutional Neural Networks (CNNs) to extract image features and transformers to understand text. The framework delivers better performance than unimodal models by combining features. While effective, the study does not include interpretability measures and does not discuss system deployment.

"Multimodal Fake News Detection" – Khattar et al., 2022 [3]. This paper suggests a multi-modal framework based on deep learning that considers both text and image properties for fake news detection. The results show higher robustness compared to mono-modal approaches. However, the system requires huge labeled data sets and involves high computational costs in its learning phase, which limits scalability.

"CroMe: Cross-Modal Tri-Transformer and Metric Learning" – Choi et al., 2025 [4]. CroMe uses a tri-transformer model with metric learning for achieving semantic-visual alignment. This technique shows high feature correlation between the modalities and high detection rates of objects. However, the model is relatively complicated and computationally expensive.

"MFND + SDML: Multimodal Fake News Detection and Dataset" – Li et al., 2025 [5]. This research presents a multimodal dataset along with an approach of deep metric learning for detecting the fake news. While this algorithm provides better discrimination capabilities on multimodal datasets, its evaluation is restricted primarily to one particular dataset.

“Bridging the Gap in Multimodal Fake News Detection” – Lin and Chen, 2023 [6]. This imbalance is corrected through pre-fusion alignment of the textual and visual feature vectors. The alignment increases the efficiency of the learning process. Nonetheless, the model is still not explainable and cannot detect misinformation at the user level.

“MCOT: Multimodal Fake News Detection with Contrastive Learning and Optimal Transport” – Wang et al., 2024 [7]. In this paper, we present a contrastive framework together with an optimal transport technique to enforce good feature correspondence between different modalities. The framework is resistant to noisy data and modality mismatch but introduces more training complexity and computation costs.

“Entity-Enhanced Framework for Multimodal Fake News Detection” – Zhao et al., 2021 [8]. This work uses NER to enrich the text representation prior to multimodal fusion, thereby improving the understanding of context and the classification performance. However, the model requires accurate entity recognition and domain coverage.

“Self-Learning Multimodal Approach for Fake News Detection” – Kumar et al., 2024 [9]. In this paper, we propose a self-learning multimodal approach to handle changing environments without full retraining. Despite enhanced practical benefits over time, the framework has stability and error accumulation issues.

“ETMA: Efficient Transformer-Based Multilevel Attention for Fake News Detection” – Yadav and Kumar, 2022 [10]. ETMA proposes a multilevel transformer attention to capture the text features in a hierarchical way. This model performance well in the text based detection, while it does not take vision or multimodal features into consideration.

“Multimodal Model for Detecting Fake News” – Roy and Singh, 2022 [11]. We fuse CNN image features with text embedding for fake news detection. It improves the detection precision over the baselines based on single modal. Yet, we have not performed enough experiments with big real datasets.

“A BERT-VGG Framework for Enhanced Fake News Detection” – Sharma and Nair, 2023 [12]. Authors make use of BERT for textual representation and VGG for image feature extraction. It shows promising multimodal performance, but the fusion strategy is simple.

“A Fuzzy-Based Multimodal Approach for Fake News Detection” – Zhang et al., 2024 [13]. This research proposes to use fuzzy logic for modeling uncertainty when classifying multimodal fake news. It adds sophistication to the decision making when uncertainty is present, but on the other hand, makes the system more complicated and requires more processing.

“NewsBag: A Multimodal Benchmark Dataset for Fake News Detection” – Khaleel and Hameed, 2024 [14]. NewsBag includes a benchmark dataset which contains samples of text and images of fake news. This helps with the data insufficiency and standardized assessment. There are no baseline detection models present in NewsBag.

“Truth Be Told: A Multimodal Ensemble Approach” – Banerjee et al., 2025 [15]. The ensemble learning has been utilized by combination of multiple multimodal classifiers to achieve better performance and accuracy. On the other hand, it leads to longer inference time and need more computational resources.

“Reasoning-Based Explainable Multimodal Fake News Detection” – Ramesh and Prakash, 2025 [16]. The framework has also included a reasoning mechanism that increases the explainability of the multimodal prediction, hence raising confidence and trust over the system. But the scalability and efficiency are issues.

“FakeNewsNet: A Data Repository with News Content, Social Context, and Spatiotemporal Information” – Shu et al., 2020 [17]. Here we present a novel and extensive dataset integrating news content with social context data. Although popular for fake news research, this dataset has limitations regarding visual diversity and time coverage.

“Fake News Detection on Social Media: A Data Mining Perspective” – Shu et al., 2017 [18]. The above classification of techniques forfake new detection based on data mining has presented to us a primary overview; However it does not provide the novel deep learning-based multimodal approaches.

“Early Detection of Fake News on Social Media Through Propagation Path Classification” – Liu and Wu, 2018 [19]. The authors have proposed a propagation based detection technique by using the structure of networks. The technique performs early detection but depends on the social graph instead of the contents.

“BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding” – Devlin et al., 2019 [20]. The paper is a foundation work introducing BERT, allowing to deeply understand context from text and thus creating the bases for most current NLP based fake news detectors. It does not however offer direct support to multimodal inputs.

Table 2.1: Existing Methods vs Proposed Methods: Comparison

Paper / Approach	**Modality**	**Model Used**	**Key Strength**	**Limitation**	**Work Advantage**
Akhtar et al., 2023	Text	ML Models	Supply chain relevance	No image analysis	Uses multimodal data
Khattar et al., 2022	Multimodal	Deep Learning	Robust classification	High computation cost	Lightweight DistilBERT
CNN + Transformer, 2023	Multimodal	CNN + Transformer	Improved accuracy	No explainability	Add AI reasoning layer
CroMe, 2025	Multimodal	Tri-Transformer	Strong feature alignment	Complex architecture	Simpler late fusion
ETMA, 2022	Text	Transformer	Strong NLP performance	No multimodal support	Combines text + image
BERT-VGG, 2023	Multimodal	BERT + CNN	Good feature extraction	Basic fusion	Advanced decision layer
Proposed System	Multimodal + AI	DistilBERT + ResNet + AI	High accuracy	Limited image training	Hybrid ML + AI Validation

2.2 RESEARCH GAP

Although a substantial body of research on fake news detection has been carried out so far, there are several fundamental shortcomings in existing research methods and real-world implementations that demonstrate a critical need for more robust, scalable and explainable systems.

· They are purely unimodal (text-only, image-only) and do not take advantage of cross-modal correlations

· Can not easily accommodate text-based and image-based misinformation combinations, thus not performing well in real-world applications

· Many multimodal fake news detection methods are opaque and they cannot explain why certain contents were flagged as fake; i.e., they serve as black-boxes.

· High computational burden imposed by complex methods hinders real-time deployment.

· They do not have any kind of reasoning capacity and do not carry out the validation of credibility, making it very difficult for them to deal with nuanced, vague, or ambiguous contexts.

· In most systems, it is not considered to involve real-world knowledge and solely relies on pattern matching of evidence.

· Many of research models have a large gap between them and deployment

· Inability to detect fake news for non-English languages.

· Limited or no multilingual support, restricting usability

· Does not offer user-based features such as Confidence or Explainability.

2.3 OBJECTIVES

The main goal of this project is to build an intelligent multimodal fake news detection system, capable of correctly classify news as real or fake using text and image features of news articles.

The goals of the proposed work is as:

· To create a text classification model based on a transformer for extracting semantic features of news headings. We'll be using DistilBERT for text extraction.

· To create a model based on CNN architecture that can extract meaningful visual features from images related to news. We'll use ResNet-18 architecture for this task.

· To develop a strategy that can effectively fuse text and image features so that we could enhance the accuracy and robustness of the fake news detection.

· To build a model based on Multi-Layer Perceptron classifier that could process fused features and output probabilities for fake and real news.

· To integrate an AI-based validation layer which uses logical and linguistic reasoning and source crediblity analysis for enhanced prediction reliability.

· To design a three-layer decision pipeline that uses ML predictions, image analysis and AI validation to provide a final cross-verified result.

· To build a responsive web application that can support interactive display of news analysis using Streamlit.

· To add support for multilingualism in the model using an automatic language detection model.

· To provide a visual confidence meter which would present the prediction probability of the news, to help us better understand the reliability of our model

· To evaluate the system using standard performance metrics such as accuracy, precision, recall, and F1-score on a balanced dataset.

Overall, the objective is to create a scalable, interpretable, and practical system that bridges the gap between research models and real-world fake news detection applications.

2.4 PROBLEM STATEMENT

The proliferation of fake and misleading news in digital media is becoming a severe problem as it affects the authenticity of information, public confidence and societal stability. In today's era of social media and online news, news could be instantaneously transmitted and thus misinformation can be easily dispersed. Manual methods for the verification process are time consuming, slow and not realistic in a real world application.

Current false news detection approaches have limitations to analyze the complexity of modern fake news which blends misleading texts with visually deceptive or irrelevant images. The former can learn the pattern of text language, but fail to exploit the semantic gap of visual features. Meanwhile, the latter cannot learn the semantic meaning of images. In this sense, unimodal models demonstrate poor performance when evaluated on realistic fake news.

Additional problems of the proposed systems are that only machine learning prediction has been used, and the semantic reasoning or confidence checking is missing. Therefore the system will not be reliable in ambiguity and gray cases. Lack of interpretability, interaction and easy deployment has been a limitation of most systems so far. Also, most systems only work for one language.

Thus there is a requirement for a solid, scalable, and explainable system that combines multimodal learning and AI-based reasoning and enable interactive human communication in real time. The proposed system addresses the requirements of the same by leveraging the transformer-based text analysis, CNN-based image analysis, and an AI-based validation layer through a 3-level decision pipe to enable a realistic, trustworthy and interpretable fake news detection system.

2.5 PROJECT PLAN

The project was carried out in a structured and systematic manner across multiple phases, ensuring efficient development, evaluation, and deployment of the Multimodal Fake News Detection system.

Phase 1: Project Initialization

· The problem statement, objectives, and scope of the multimodal fake news detection system were clearly defined and finalized.

· An extensive literature survey was conducted on fake news detection, multimodal learning, CNNs, transformer models, and content verification systems.

· Identified the limitations in present unimodal and multimodal approaches and accordingly research areas were found.

· Design the architecture, finalize the implementation method, dataset suppliers, the tools to be used like Python, PyTorch, Hugging Face Transformers, OpenCV and Streamlit.

Phase 2: Dataset Collection and Preprocessing

· Available public multimodal fake news datasets with news text and corresponding images were collected and curated.

· Preprocessing of text such as cleaning, tokenization, padding, normalization was carried out.

· Image preprocessing such as resizing, normalization, format standardization was done.

· Split dataset into training, validation, and testing sets to have reliable results.

Phase 3: Model Design and Development

· DistilBERT a transformer based model was used to extract contextual and semantic information from text data.

· ResNet-18 a CNN based model was used to extract discriminative features from images.

· Both text and image models were first trained separately to compare modality wise performances.

· Performance and consistency of the model were verified on different samples.

Phase 4: Evaluation and Optimization

· Models were validated through common metrics like accuracy, precision, recall and F1 score.

· The performance was then measured for the true and fake news.

· Hyperparameter tuning, regularization and optimization methods were used to enhance the model's ability to generalize to unknown data.

· The benefits of multimodal fusion were then compared to unimodal performance.

Phase 5: Multimodal Integration, AI Validation, Deployment, and Documentation

· Text and visual features are merged using a late fusion technique to create a combined multimodal representation.

· An MLP classifier was built to predict final prediction probabilities.

· An AI-based validation layer was used to perform semantic inference to boost prediction accuracy.

· Build a 3-layer pipeline combining ML prediction and AI validation.

· Build a Streamlit web application to allow real-time fake news prediction with interactive visualization.

· More advanced features were implemented, such as multilingual support, a confidence meter to display the level of confidence, and a session-based analytics display.

· The system was tested, documented, and ready for final presentation and evaluation as well as deployment.

CHAPTER 3

TECHNICAL SPECIFICATION

3.1 REQUIREMENTS

3.1.1 Functional Requirements

The functional requirements state the functions that are expected to be delivered by the system:

· Accept input of text of the news headline and the image url and the domain of the source from the user.

· Preprocess text inputs such as tokenization, normalization and cleaning.

· Preprocess image inputs such as resizing, normalization and file format transformation.

· Classify the texts by transformer-based models (DistilBERT).

· Extract features of images by Convolutional Neural Network models (ResNet-18).

· Combine text and image features through multimodal fusion.

· Generate fake and real news prediction probabilities by an MLP classifier.

· Incorporate an AI-based validation layer to carry out semantic reasoning and truth checking.

· Show the classification result, the confidence score and the explanation of why it is the classification result.

· Supports multiple languages input with automatic language detection.

· Displays a visual confidence meter of prediction.

· Saves the session-based statistics and history of recent detections.

· Allows independent run of text or image models.

3.1.2 Non-Functional Requirements

Non functional requirements describe Quality and Performance aspects of a system. The non functional requirements of the system in scope are given below:

· Accuracy: Quality of the prediction of the system should be acceptable and consistent.

· Scalability: Ability of the system to increase input data volume while not compromising the prediction performance.

· Usability: The system interface should be user-friendly, interactive and easily understandable.

· Performance: A system response time must be of acceptable nature.

· Maintainability: Design of the system should be easily modifiable.

· Reliability: System performance should not change during its normal use.

· Security: Data fed in the system should be secure.

· Interpretability: The prediction should be easily understandable and accompanied by measures of confidence and explanations.

· Accessibility: The system must support multilingual inputs.

3.1.3 Dataset Requirement

The effectiveness of the suggested Multimodal Fake News Detection system heavily relies on the quality and organisation of the dataset. Because the system deals with two modalities namely text and image, the dataset must be appropriate for multimodal learning i.e., containing pairs of news content and image which belong together.

The dataset used in this project is a corpus comprising of news with both text and images. The dataset contains 9,000 news samples, divided evenly between fake and real news samples so that the training can be done efficiently without biased outcomes. The samples have been divided into three parts namely train, validation and test. For each data point, we are provided with news headline, cleaned text, image url and domain source with label as whether the news is fake or real. The textual part is fed for the semantic features using a model like distilBERT, the images part is fed to learn visual features using models like ResNet-18.

The dataset is collected from publically available news collections which are Reddit based and covering various fields ranging frompolitics to science and other. The varied news coverages enhance the model with generalization capability among different types of content and real world situations. The effective alignment among text and images ensures the efficient multimodal feature learning.

In the summary, this dataset seems to be a good one in terms of its structure, balance and variety.

3.2 FEASIBILITY STUDY

3.2.1 Technical Feasibility

The project is technically feasible because of the maturity of existing AI frameworks and resources available.

· Availability of pre-trained models (transformer models, like DistilBERT) for NLP-based solutions.

· Availability of established CNN architectures (like ResNet - 18) for image recognition tasks.

· Availability of established and user-friendly deep learning frameworks (like Tensorflow and Pytorch).

· Availability of high performance, cloud-based, GPUs for training complex models.

· Availability of open-source data and tools that allows for reproduction of the results.

· Availability of fast, high-performance, AI inference APIs to be used for validation at reasoning level.

3.2.2 Economic Feasibility

Economic viability of the project: It's economically viable, mainly because it is dependent on free/open source tools and public resources:

· Use of free and open-source libraries to develop.

· Use of publically available datasets to test.

· low hardware cost using computation through cloud computing.

· No license costs for development tools.

· Inexpensive to deploy using lightweight web framework.

3.2.3 Social Feasibility

System is socially viable and will offer many benefits to society, in the fight against misinformation:

· It aids in the limitation of misinformation and false content.

· SIt promotes informed decisions to users and people online.

· It will encourage digital literacy and promote safe use of information online.

· It assists fact-checkers and researchers in verifying content.

· It instills transparency and trust in digital platform media.

3.3 SYSTEM SPECIFICATION

3.3.1 Hardware Specification

The hardware requirements for development and execution include:

· Processor: Intel i5 or equivalent (minimum)

· RAM: 8 GB (minimum), 16 GB (recommended)

· Storage: 256 GB SSD or higher

· GPU: NVIDIA GPU with CUDA support (optional but recommended)

· Internet connectivity for dataset access and cloud execution

3.3.2 Software Specification

The software requirements used in the project are:

· Operating System: Windows / Linux

· Programming Language: Python 3.x

· Deep Learning Frameworks: TensorFlow, PyTorch

· NLP Libraries: Hugging Face Transformers, NLTK

· Computer Vision Libraries: OpenCV, Pillow

· Web Framework: Streamlit (for user interface)

· API Integration: AI inference API for semantic validation

· Development Environment:VS Code, Google Colab

CHAPTER 4

SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

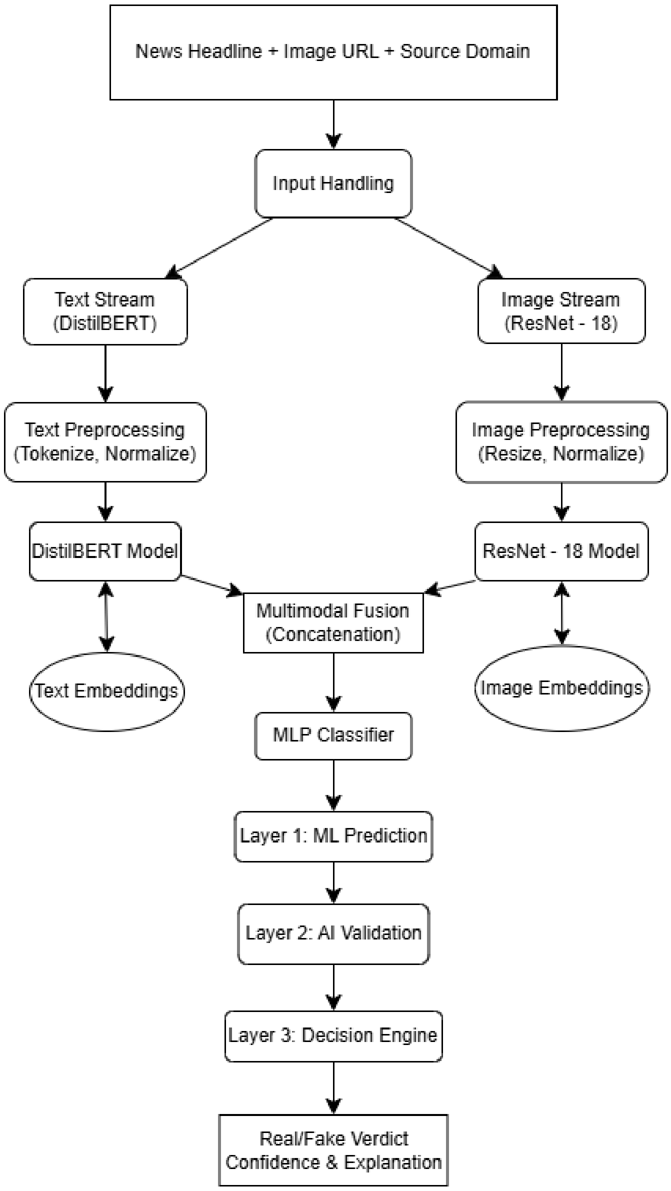

Figure 4.1: System Architecture Diagram

The presented system shown in Figure 4.1, proposes a modular modal approach, performing detection of fake/misinformation using the joint analysis of text and image modalities alongside an AI-based semantic validation mechanism. The structure of this design stems from the requirement to deal with unimodal system's drawbacks and is based on the goal to improve both prediction accuracy and interpretability and usability at real time

It takes three basic inputs: the news headline, the URL of the image and the source domain. These inputs are fed to the system through two pipelines in parallel: text pipeline and image pipeline and then a fusion process and a multi-layered decision system.

In the text processing pipeline, the input headline goes through regular processing stages such as tokenization, padding and normalization so that the textual data can be accepted by the deep learning model. The tokenized text will then feed into a transformer based architecture which, in this case, is the DistilBERT which is known to represent deep semantic information and contextual relationship of the text. And this process outputs a rich text embedding that could be regarded as a dense vector.

Meanwhile the image processing pipeline takes charge of the visual element by fetching the image from the URL and preprocessing operations such as resizing, normalization are performed on it. The processed image is then fed to the Convolutional Neural Network, ResNet-18. A set of high level discriminative visual features is extracted using the CNN. The image embedding contains the visual features of the image, which might serve as evidence of manipulation, irrelevance to the given text or inconsistency.

Outputs from both streams are combined using a multimodal fusion layer through a late fusion approach by concatenating the embeddings from the text and the image to create a single feature vector. This combined vector then passes through few fully connected layers where dropout regularization is employed to over-fit. The classification is done by a softmax output layer giving probabilistic values as the probability for fake news and real news. This is layer 1 of the system (Machine Learning Model).

This second layer- the AI reasoning-based validation module - was added to boost confidence and provide more reasoning-based output to compensate for the shortcomings of a fully data-driven approach. This layer works by performing semantic reasoning, language processing, and checking the validity of sources, providing its own verdict and clear explanations as well. The integration of a reasoning engine into the system makes the explanations more robust.

The final decision is made based on the 3rd layer which is named Decision Engine

(Layer 3), taking the prediction result from machine learning model and AI validation module into consideration. Where machine learning model and AI validation module agree, the final decision would be deemed as a high confidence prediction. In the opposite situation, the decision is made on basis of having the higher confidence signals or more strong reasoning signals between both sources, which is expected to yield a robust result.

The system finally generates a full output containing: predicted class(real/fake), confidence associated, explaining reasoning and additional values such as trust badges and red flags. Further from the main purpose, this architecture enables also additional features such as support of multiple languages and confidence display as well as user real-time interaction through web application implemented with streamlit.

The modularity of the architecture allows testing and refining each part of the system individually and improves the robustness of the whole system thanks to several stages of validation, and flexibility for subsequent developments like enhanced fusion methods, explainability components or adaptivity mechanisms.

4.2 DETAILED DESIGN

This system is strongly based on the concept of modularity, scalability and flexibility, with the goal of making development, evaluation and extensions systematic and easier. Each component of the processing pipeline has been created as a separated module which makes every component removable, replaceable or upgradeable without breaking the rest of the system. In addition to debugging and experiments, it would make modifications or addition of new models or procedures very fast.

The architecture is composed of two parallel streams, a text stream and an image stream. The separation enables efficient and easy learning, training and testing of the features for each modality independently. The text stream is implemented using a transformer based DistilBERT model, enabling deep semantic and contextual understanding of the news headlines and so capable of identifying the deceitful and ambiguous textual data.

Concurrently, the image stream adopts a Convolutional Neural Network,namely ResNet-18, for the extraction of discriminative visual features from the given images. It learns higher-level visual representations capable of distinguishing incongruous images, manipulations, or irrelevant images, which usually lead to fake news.

The multimodal fusion layer is the main processing unit of the network. Textual and visual embeddings are merged using a late fusion (concatenation) technique so that semantic and visual features of the input can be efficiently exploited in prediction. The obtained fused feature vector is then fed into a MLP for learning non-linear correlations between modalities and generating the probability distribution on class labels. A dropout mechanism is implemented in the network to help improve the generalization performance of the model and avoid over-fitting.

For a higher prediction accuracy and a better interpretability of results, a further validation layer powered by AI is added. The validation layer consists of semantic reasoning, linguistics analysis and source authenticity check and it provides an independent assessment of the information to be analyzed. The system, combining a data driven model with a reasoning based validation, reaches an equilibrium of prediction accuracy and trustworthiness.

The model's output along with AI validation layer's output feeds into the decision engine to establish a three layer prediction pipeline. This layered approach helps in being robust by providing cross validation between different ways of reasoning which would reduce chances of wrong prediction.

In terms of usability, the system facilitates live usage via the web-interface that is implemented with the Streamlit framework. The system also includes features to accommodate multi-language inputs, present the confidence score of the input, and support session based analytics to ensure a smooth live deployment of the system.

Overall, the modular design can facilitate a high extensibility of the system when new breakthroughs are made. It is conceivable to have further extensions such as more sophisticated fusion methods, explainable modules, real time data feeding, or even production-level large scale deployment.

4.3 DATA FLOW DIAGRAM

4.3.1 Level – 0

In the Level-0 DFD, the Multimodal Fake News Detection system is considered as a single monolithic process that interacts with an external entity (the user). It aims to give

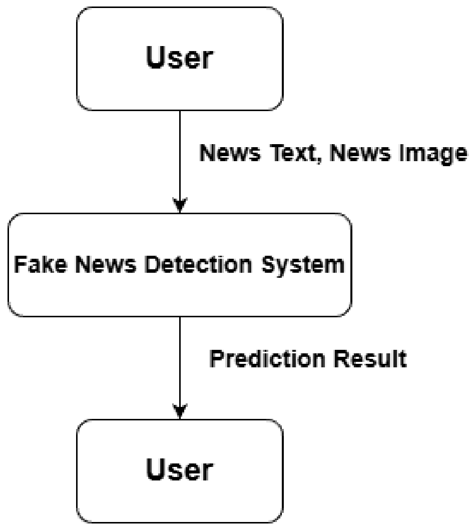

Figure 4.2: Level-0 Data Flow Diagram of the Proposed System

a view of the system boundary without disclosing its internal implementation details. At this level, the user is considered as the single source of input that provides data of type News Headline, Image URL, and Source Domain. These data elements comprise the multimodal nature of the inputs and are considered as a single input to the system. The system then consumes this input without exposing the process in between.

The system processes the received inputs by carrying out the internal processing using its built-in pipeline which works on the multimodal aspect. The inner process is not specified but involves text processing, image processing, and decision-making algorithms. At this level, the focus is on what the system does rather than how it does it. After performing internal processing on the input data, the system returns a single output of type Prediction Result that states if the news is real or fake. Additionally, the system also returns a Confidence Score indicating how confident it is on the result, and also Reasoning for it.

The output obtained is then sent back to the user which completes the cycle of interaction.

4.3.2 Level – 1

Level-1 Data Flow Diagram illustrates the internal pipeline processing focusing on the machine learning-based multimodal system design. Different from the system in Level-0, here, the system is decomposed into several functional parts and shows how data is moved among different modules. The input is received from user, which is then

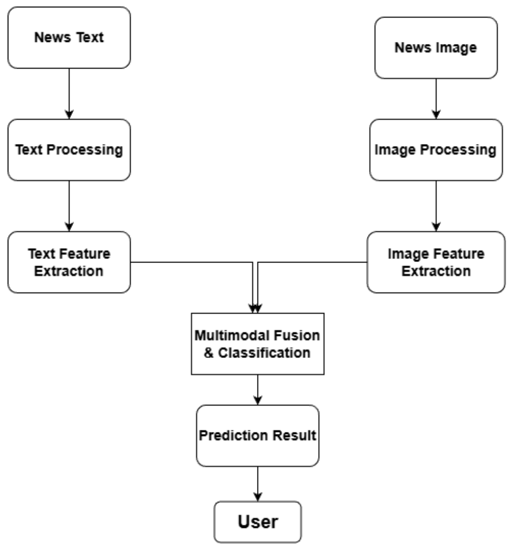

Figure 4.3: Level-1 Data Flow Diagram of the Proposed System

segregated into two parallel paths, text path and image path, and both modalities can be processed independently using domain-specific models to obtain relative features.

Within text path, the text news headline needs to be processed with techniques like tokenization, normalization, and padding to meet requirements for deep learning model. After processing, DistilBERT is used to learn contextual embeddings. It extracts semantics of text based on the relationships of words in the text context. Similarly, in image path, input image will be preprocessed, e.g., scaled and normalized, then ResNet-18 is applied to extract high-level image features as a feature vector of fixed length.

After receiving text embedding from DistilBERT and image feature from ResNet-18, text and image features are combined in a multimodal fusion module. We adopt late fusion technique to concatenate these two embedding vectors to form the final feature vector that contains information of both modalities, so the fused vector represents more comprehensive features.

Then the fused features are input to a Multi-Layer Perceptron (MLP) classifier, and then the probability-based predictions can be obtained that reflects the confidence in predicting the news as fake or real news. This represents the core machine learning prediction process.

4.3.3 Level – 2

The Level-2 DFD visualises the AI driven validation and decision making part of the Multimodal Fake News Detection system. The machine learning prediction provided

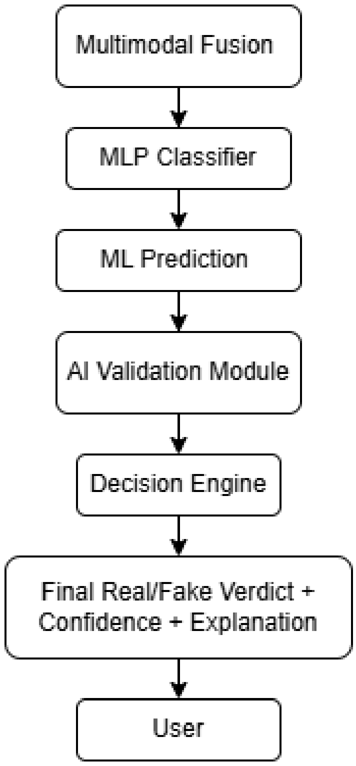

Figure 4.4: Level-2 Data Flow Diagram of the Proposed System

by the ML model at Level-1 cannot be solely used to rely on the result generated by it, as there are some circumstances under which the predictions are uncertain and less trustworthy and interpretable. So the Machine learning prediction generated at Level-1 is fed to the AI validation module which analyzes based on the semantic reasoning, language pattern analysis and source reliability of the input news text, makes its own decision and gives explanations. The decisions and explanations are then fed to the Confidence evaluation block where machine learning output and AI validation output is compared with their respective confidence values to decide the final outcome. Decision engine at Level-2 is responsible for making decisions based on agreement and dispute between the outputs from machine learning model and AI validation module. High confidence will be awarded when the system decision is agreed upon by both machines and explanations are justified. However, when the outputs are disagreed, the output from the confident model (either ML model or AI validation model) and its relevant explanations are chosen. The final output would also involve classifying the news as fake or real, its confidence level and explanation. Level-2 proves that this system is not just based on Machine Learning model predictions, rather is a combination of both Machine Learning models with AI based validation.

4.4 CLASS DIAGRAM

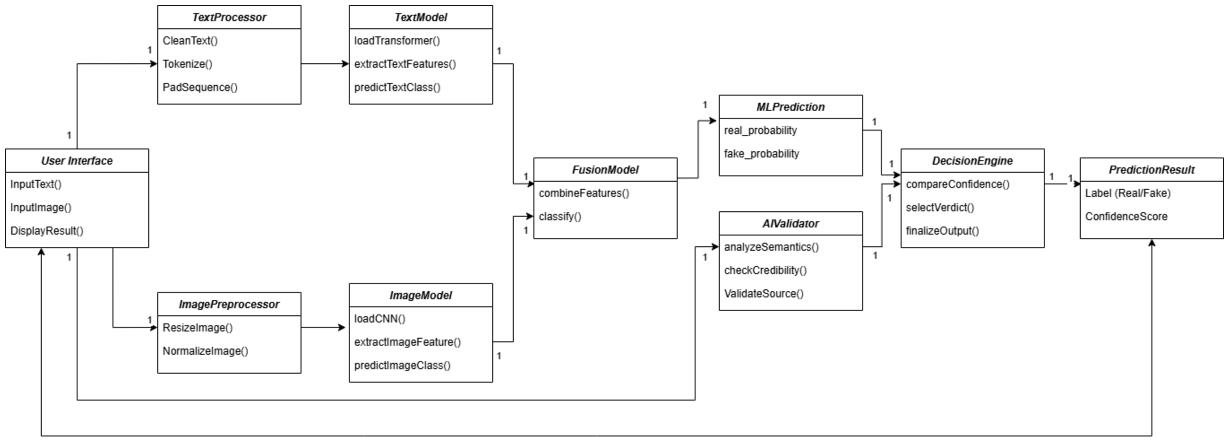

Figure 4.5: Class Diagram of the Proposed System

Class diagram explains the logical and modular design of the Multimodal Fake News Detection system. It also explains the relationship between various classes which are designed to process the data, extract the features, and perform prediction. The system is designed in a pipeline-oriented and modular approach with each class performing a specific function, thereby improving clarity and ease of maintenance.

The UserInterface class is the basic interaction level of the user with the system. It is used for taking user's inputs such as headline and image through the methods like InputText() and InputImage() and also to display the final predicted value through the DisplayResult() method.

Firstly, the textual data is processed by the TextProcessor class, which carries out fundamental preprocessing actions like CleanText(), Tokenize() and PadSequence() on it so that it can be used in further step of feature extraction after preprocessing. Then, the processed text data is forwarded to the TextModel class, where it is processed by the transformer based model named DistilBERT, it is capable of extracting the contextual and semantic information, from the method loadTransformer(), extractTextFeatures() and predictTextClass(), we can obtain the significant text embedding. In parallel, the visual data is processed by ImagePreprocessor class, it carries out the operation of ResizeImage(), NormalizeImage(), etc, to make sure the image data can be used to the model without error. Then, the preprocessed image data is forwarded to the ImageModel class where the image data is processed by CNN (ResNet-18) to extract high-level visual feature through the method loadCNN(), extractImageFeature() and predictImageClass().

In the FusionModel class, it takes the text and image model as inputs and is the central class that connects them. The combineFeatures() function combines the embeddings from both text and image models into a single feature vector (late fusion). The classify() function then uses this combination to produce the machine learning prediction. The MLPrediction class takes in the combined probability for both classes (realprobability and fakeprobability) to give an indication of how confident the model is, these then act as inputs to another level of validation.

To increase reliability, there's the AIValidator class which carries out a separate verification of the input information. It employs semantic reasoning, analysis of linguistic features and source credibility by using a combination of methods such as analyzeSemantics(), checkCredibility() and validateSource(). This module injects reasoning into the analysis of the text in addition to the predictive machine learning model.

The DecisionEngine class is the concluding decision-maker in this workflow. It aggregates and cross-references the output of the MLPrediction and AIValidator components by utilizing compareConfidence(), selection of the best verdict using selectVerdict() and finishing up the prediction by finalizeOutput().

This layered system enables predictions which are more robust. The PredictionResult class holds simply the results of the predictions i.e. The label (true or fake) and the confidence level associated with it, and passes the values to the UserInterface. The class diagram above thus shows the hybrid system architecture, with multimodal ML and an AI decision and validation system. The system's modular architecture ensures that it can be scaled and maintained easily, and makes the system as a whole more modular.

CHAPTER 5

METHODOLOGY AND TESTING

5.1 METHODOLOGY

The implemented Multimodal Fake News Detection has a systematically engineered and layered pipeline system where text and images are jointly processed to make a prediction. During the design stage this system was built with an objective of scalability, extensibility and interpretability so that each layer can be evaluated independently, optimized or extended. Such a process of using an individual module directly results in positive effect during experimenting, debugging and enhancing the system.

The system starts with the User Interface Module as the first point of the system's interaction with the outside world. It serves as an input interface where the system receives values from the user (i.e. A news headline, the URL for the image and the source domain). Apart from inputs, the user interface can also support outputs that offer relevant results such as the classification obtained, a value indicating confidence and an explanation for the obtained classification.

After taking inputs from the user, the system passes it through two parallel pipelines namely the Text Processing Module and the Image Processing Module so that both of the modalities are processed in separate ways.

The Text Processing Module performs feature extraction of linguistically meaningful terms from the news headline. The original input text is pre-processed using text cleaning, tokenization, padding and normalization procedures so that it is well-formatted and suitable to be provided as input to deep learning models. Then, it is fed to transformer based model, DistilBERT which understands deep semantic relations, dependencies among words and context that helps the system to identify manipulative, deceptive, or ambiguous statements in news. The output from this module is a compact text embedding which represents the semantics of the input headline in a high-level feature vector.

Concurrently, the Image Processing Module deals with the visual aspects of the news. The system extracts the image by fetching it from the URL and preprocesses it with resizing, normalization and format correction such that it fits into the model's requirements. The normalization processes allow to provide to the model a good and stable quality input. The resulting image is finally pushed into a Convolutional Neural Network model, ResNet-18, which will extract the visual features. This type of model is a suitable method to detect relevant features such as tampered images or irrelevant pictures, and such as the dissimilarity between image and the relevant article. The resulting image embedding will sum up these features with a unique and compact vector.

After features are extracted, both the embeddings are placed into Multimodal Fusion Module, where it stands as the heart of the system. We adopt late fusion (concatenation) approach and the text and image embeddings are simply concatenated together into one feature vector, to retain the semantics and visual properties and for them to together contribute for classification. The combined vector is then feed to a MLP classifier for learning the relations between both modalities and it comprises fully connected layers, non-linear activation functions and dropout regularization to prevent the model from overfitting and provide good generalization. The output layer uses a softmax function which predicts probability based prediction of whether the news is fake or not. This forms Layer 1: Machine Learning Model.

In order to overcome the disadvantages of exclusively data-driven models and improve the level of interpretability, another AI Validation Module has been included. It operates separately of the neural network and uses semantics reasoning, language pattern matching, source credibility analysis. It uses the headline of the input and context of domain to check consistency, weird words, low-credibility domain, etc. Differently of machine learning model that looks for patterns that have been learned previously, this layer provides a layer based on reasoning and provides an additional validation result and an additional rationale. Thus, we obtain Layer 2: AI Validation, which improves drastically the possibility to justify the prediction.

The output from the two layers is passed to the Decision Module (or Decision Engine), Layer 3. Layer 3 is essential in ensuring that the final prediction is stable. This is done by comparing outputs, on a number of basis's, such as the strength of the supporting evidence, the agreement between the outputs, and the confidence score attributed to the output from the system. If the machine learning, and AI validation module agree on a prediction then a high confidence score will be assigned to the prediction. If the two disagree then the decision engine will decide which output is better supported, and that prediction will be selected.

The last part the system produces is the actual output which would consist of the classification (fake or real), the confidence level, and a thorough explanation of the classification. These results are then delivered to the user via the interface and would help the user to make a knowledgeable decision about whether the source is fake or not. Other indicators such as potential red flags and credibility indicators might also be introduced into the output to increase the system's comprehensibility.

In a nutshell, this modular design has achieved the fusion of multimodal deep learning and AI based reasoning to achieve a system for the effective, efficient and interpretable fake news detection system. Moreover, this design is extensible enough to incorporate future techniques, for example, advanced fusion strategies, XAI models, streaming data, or distributed system frameworks.

5.2 TESTING

System testing is carried out in a hierarchical and exhaustive fashion in order to ensure the correctness, completeness, and robustness of the various components of the proposed system and also of the system as a whole. Since the proposed system is modular, tests are run at different stages of the pipeline to validate the working of each individual module, and of their interdependence on one another.

On the most fundamental level, the unit testing is applied to each module, such as text preprocessing, image preprocessing and feature extraction modules. Each module is tested individually, and should be validated as performing its operations accurately and producing the correct outputs given the correct inputs. As an example, for the text preprocessing module, we are concerned that it performs correct tokenization, padding and normalization and so on, whereas, for image processing module we are concerned with that it does resize and normalize images as needed, etc.

After the unit test is performed the model is tested for its efficiency for the ML models in. We used accuracy, precision, recall and F-1 score to quantify the performance of models. We test both text based and image based model (DistilBERT and ResNet-18 respectively) separately so as to assess the performance of models for specific modalities before they are combined to make a multimodal model.

After validating the single models, the models are integrated by integration testing where we check that all the different modules communicate with each other as intended, such as the flow of data starting from the preprocessing to the feature extraction stage, then fusion and classification. Especial care is given to the multimodal fusion stage whentext and images embedding are merged, ensuring the features are aligned, the dimensional consistency and that the concatenations occur properly, which contributes to the prediction.

Other than the machine learning part of the system, we have also tested the AI validation module on its own. The function of the AI validation module is to evaluate its ability to reason semantically, analyze patterns in languages and analyze the credibility of a source. The output from the validation module was compared against the output from the machine learning part in order to test its consistency and reliability as well as see how it complements the performance of the system.

At the top level, system testing takes a varied input data-just text inputs, just image inputs and fully multimodal inputs- to check the pipeline against the most diverse conditions. The consistency in system behavior among the different input configurations is monitored in this level to test the fault tolerant capacity of the system.

Last but not least, the user-level testing was performed by the web interface developed using Streamlit. This testing part ensures user interactions such as inputs receiving, response speed and outputs display are user friendly and convenient to use. Input features such as data input, confidence scores, and explanatory insights displayed on the interface are examined and checked.

Through combined use of these testing strategies, a holistic validation strategy has been developed to ensure the system is robust and predictable in varying conditions. Together, unit testing, model testing, integration testing, system testing, and user-level testing results in a system that has both high accuracy, high robustness, and high usability.

CHAPTER 6

PROJECT IMPLEMENTATION

Multimodal Fake News Detection System is implemented using python programming language with contemporary deep learning models and web frameworks. Our aim is to make an application that is efficient, scalable and user-interactive. It is implemented using a modular design where each module's function is coded separately and then connected with each other to form a pipeline. This makes the development and debugging of each module easier, while also allowing modifications to upgrade the model.

At the center of the system is PyTorch, which is used for constructing, training, and implementing the models of neural networks. When analyzing text, the system uses Hugging Face Transformers, more particularly DistilBERT model to capture the context and semantics of news headlines. As DistilBERT is a lighter version but a powerful transformer model, it achieves good performance with less computation power. The input text is then fed to DistilBERT tokenizer to perform tokenization, encoding and padding to a determined max sequence length to prepare input for the model.

The image processing pipeline is written in torchvision, using a pre-trained ResNet-18 as visual feature extractors. To make the network used for feature representation instead of classification, the last fully connected layer in ResNet-18 is altered, so that the model will return a fixed dimension vector as an embedding vector. The preprocessing part (resizing, normalizing, converting format, etc.) is carried out by OpenCV and Pillow.

The multimodal fusion and classification module is implemented as a custom neural network in PyTorch. The dimension-reduced text embeddings(768-dimensional) and image embeddings(512-dimensional) are concatenated into a feature vector, containing semantic and visual features. The combined feature vector is fed into a Multi-Layer Perceptron(MLP) consisting of stacked fully-connected layers with ReLU activation and dropouts for preventing over-fitting. The output layer takes softmax function which outputs the classification probability on fake news and real news. This is the main ML inference layer.

To support the data driven predictions, the system uses an additional AI based validation layer implemented via an external AI inference API, which is intended to analyze the inputs at a higher level. This module takes into account semantic understanding, evaluation of linguistic trends, analysis of source credibility etc based on input headline and domain. This layer is different from the neural layer as it is a reasoning based layer, not a learned representations based layer and offers a separate, transparent verdict, together with the relevant reasoning.

Finally, a decision engine is employed to merge the results of the machine learning model and AI validation module. In doing so, confidence scores, layer concordance and reasoning strength are evaluated to ultimately make a prediction. The confidence assigned to a prediction will be enhanced if both the machine learning model and AI validation module return the same prediction. If the machine learning model and AI validation module return conflicting predictions, the decision engine should use its judgment to favor the prediction with better supporting evidence to enhance system reliability.

For the user interaction, an interface is created using Streamlit where users can type their news headline, enter a news image URL, provide the source domain and receive prediction in real-time through the web interface. Once the input is provided by the user, the data goes through processing and finally a prediction along with confidence, explainability and a confidence meter to clearly illustrate the confidence score for a better user interface.

Overall the implementation integrated multimodal deep learning methods and AI-based validation mechanisms seamlessly, presented as a simple and intuitive application. In addition to providing reliable real-time detection of fake news, the system can offer meaningful explanation and is thus useful for both deployment and improvement of research in the future.

6.1 DATA PREPROCESSING

The input data has to be pre-processed so that it is clean, consistent and acceptable to be used efficiently in model training. As the system we have developed handles both the text and image data independently, preprocessing stages are different but aligned with each other. For textual input data, a series of operations including deleting extra characters, changing cases to lowercase, and removing unwanted information (special symbols, noise) are performed. The clean text is then fed into the tokenization process where it is segmented into various tokens to be processed. To ensure that all input sequences have same length, padding and encoding steps are taken which also helps model in understanding the semantic relationships rather than insignificant deviations in the data.

The preprocessing stages for visual data will convert all images to a single dimension, and perform normalization over pixel values and convert the images to the suitable formats required by deep learning networks. These stages can ensure the similarity of all input images and the quality of feature extraction by deep learning models (like ResNet-18). Meanwhile, abnormal and unreachable image URLs must be handled separately.

In summary, the data preprocessing can enable model to receive similar input images and text features, which enhances the model stability, performance and reliability greatly.

6.2 TRAINING STRATEGY

The training approach ensures the model effectively learns from the dataset while having good generalization capability. The dataset is divided into train set, validation set, test set, and trained on one portion and tested on the unseen samples to make sure the estimate on model performance is unbiased. During training, multimodal model learn from the textual and visual inputs in parallel. In the training process, predictions are made in the forward propagation phase, then the errors are computed in the backward propagation phase and used to update model weights. Optimization algorithm like Adam is used to minimize loss function, making the prediction accurate over subsequent epochs. Some regularization methods like dropout is adopted in the neural network to avoid overfitting and make sure the model not memorize the train set, instead, learn the generalization patterns and generalize to unseen inputs. All performance metrics are monitored in the process of training including accuracy, precision, recall, and F1-score to have a parameter tuning for the best model.

6.3 LIMITATIONS

Although accuracy and performance are quite high in the developed system there are some limitations which need to be noted. One of the limitations of the system is its dependence on the quality of the input data provided. If the text content is vague, or the image content is not available or is irrelevant then the system's prediction accuracy is compromised.

Another limitation is its dependency on an external AI based validation API, this can lead to delay and dependency on external factors. If the external API is down or is slow, then it will impact the system performance.Also, the system achieved promising results on the dataset, it may not be so effective when used on newly or previously unseen forms of fake news if they are not very close to the training data, it means it should be updated and retrained in order to catch the fake news form evolving trend.

In terms of fusion, the present fusion method, the simple feature concatenation may limit the system on realizing more complex relationship between text and image, some advanced fusion methods could perform even better.

In a word, recognizing the above shortcomings gives good hints on how to enhance the system in the future.

CHAPTER 7

RESULTS AND DISCUSSION

7.1 RESULTS

The proposed Multimodal Fake News Detection system is evaluated on a balanced dataset comprising 9000 news instances, with fake and real news equally represented in terms of quantity. Training, validation and test sets were created to facilitate accurate analysis of the system's learning and generalization ability.

Test Accuracy 98.63% F1-score 98.63% is found to be quite high showing excellent classification performance over both classes. The validation accuracy raised up to 98.43% proving the model does not heavily overfit the training data and is able to predict on unseen data well. The precision and recall is seen to be high across both classes. The precision values reach nearly 99% meaning that this is excellent for eliminating False Positives and False Negatives which is essential when dealing with fake news.

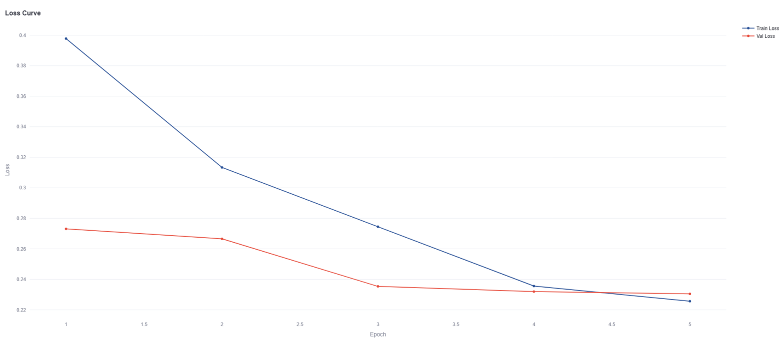

Figure 7.1: Loss Curve of the Model

[INFO] Captions
Rule: Caption numbering gap: expected 5, found 7
Expected: 5
Detected: 7
Confidence: 100.0%

We show the training behavior of the model by training loss an Figure 7.1, displays that the loss of training set and validation set decrease across epochs; the training curves of loss show no significant fluctuations during the training procedure, and its accuracy increases as time goes by, which signifies that the model is stable in training and the optimization works well. Moreover, the trend of training and validation loss curves do not separate obviously, showing the model both can fit the training data and generalize to unseen samples successfully, which proved that regularization and architecture design are helpful to avoid over-fitting. Analogously, Figure 7.2, shows the curves of the training and validation accuracy, where they both

Figure 7.2: Accuracy Curve of the Model

demonstrate a monotonic increasing behavior that converge toward an almost optimum accuracy. The similarity of training and validation accuracy, once again, supports the validity and generalization of the model. Overall, the two curves suggest the well-trained status and high performances without fragility. In Figure 7.3, the complete

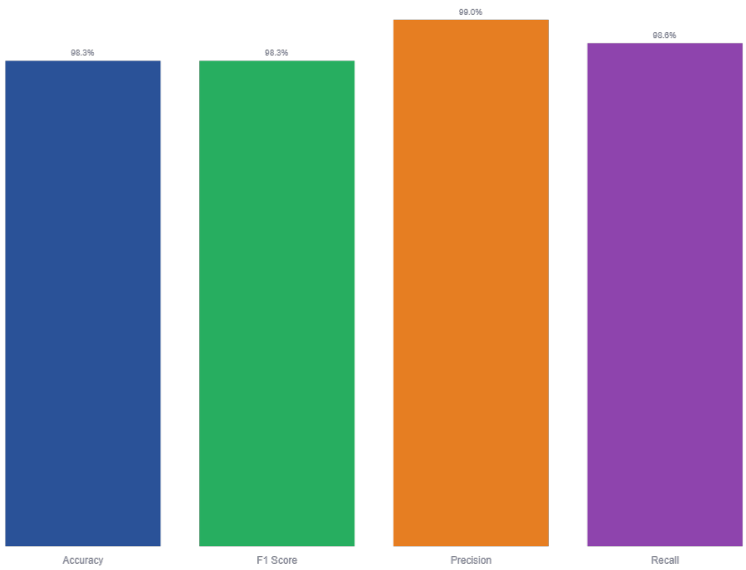

Figure 7.3: Overall Performance Evaluation of the Model

evaluation measurements of the model have been compiled which represents the accuracy, precision, recall and F1-score of the model. The above metrics is greater than 98%, which suggests that the model is good at classifying both the real news and fake news. It means that the model is balanced among all the metrics, and its performance is reliable.

Besides the system performance, components were individually assessed. The text-based model based on DistilBERT shows high performance, which can catch the context and semantic information of news headlines. That characteristic helps in detecting linguistic subtleties often found in deceptive articles. In comparison, image-based model based on ResNet-18, can successfully extract the visual features of images, then the inconsistency between image and text can be detected by the system.

To also gauge robustness, the system was subjected to inputs consisting of a single modality as text-only input, image-only input, and a multimodal input combining text and image. It can be seen that all modalities perform acceptably on their own and even better when combining both modalities of text and image, further indicating the advantage of the multimodal strategy as the two modalities supply different pieces of information to supplement classification.

Overall, the experimental outcomes above reveal that our designed system achieved outstanding performance with high accuracy, excellent generalization and stability under varying environments. By adopting deep multimodal learning along with intelligent verification assisted by AI, our system can perfectly deal with fake news detection in a practical and appropriate application where trustworthiness as well as explainability are extremely desired.

7.2 DISCUSSION

This experimentation thoroughly shows the benefits of merging a transformer-based text analysis with a CNN-based image analysis system within a multimodal model. The sustained high levels of accuracy demonstrate that textual contextual information appears to be the predominant factor in fake news detection. Specifically, the fact that the text based modality performed so well when using DistilBERT signifies its capacity to extract sophisticated textual relationships, false descriptions, and semantic irregularities found in fake news.

However, the image based component which was developed using ResNet-18 is very significant in the improvement of the system's robustness. Even though text features provided a good baseline performance, the use of visual information in addition to text helps the system detect the inconsistencies across modalities e.g. Inconsistencies between the images and the headlines. This synergy across modalities increases the robustness of the system to be applicable in the real-world where fake news makes use of deceptive or irrelevant images to fool the users.

The inclusion of AI-based validation layer is a key feature in the described approach as it provides a second line of defense. While the main machine learning model is based on learned representations, the AI-based layer conducts logical reasoning, checks linguistic patterns and analyses source veracity. By carrying out its own analysis of the input text, it presents its conclusion with detailed explanations that would add certainty to predictions and a better insight. The layered approach will significantly lower chances of incorrect classification.

The flexibility of the system can also be observed from its robustness when processing inputs of different modalities: text-only, image-only and multimodal images/texts. The system still works when one modality is absent, hence it can be useful for applications where modalities are not available at all times. While it performs best with multimodal images/texts, this also validates the benefits of multimodal learning.

From the aspect of usability, the web interface developed by using Streamlit has greatly improved the user's experience, where the interface can receive the user's input in real time and return the results, and the display of confidence scores, analysis explanations, and graphical indicators also strengthens the user's understanding and trust in the system.

Despite the excellent results achieved, some points are worth highlighting for future improvement. The system heavily relies on the quality of the input information and thus noisy or incomplete inputs will probably result in a reduction of accuracy. Furthermore, although the currentlate fusion approach performed well, additional, sophisticated fusion methods may potentially further improve the prediction based on deeper inter-modal dependencies.

In summary, the results of our proposed system demonstrate the effective integration of multimodal deep learning with AI-driven verification, enabling us to develop a solution that is both accurate and robust, and also transparent and usable. The proposed system addresses some of the drawbacks of conventional unimodal systems, indicating its promising potential for real-world application in fighting the circulation of false information.

CHAPTER 8

CONCLUSION AND FURTHER ENHANCEMENTS

8.1 CONCLUSION

This paper is about an efficient and reliable Multimodal Fake News Detection system that leverages sophisticated deep learning techniques and AI-based reasoning to cope with the escalating problem of fake news. The proposed system fuses advanced transformer- based NLP and CNNs based image analysis to capture the essence of semantic and visual information of the news data. It utilizes DistilBERT to perceive the meaning of text thereby spotting inconsistencies and malacious patterns in the text while it utilizes ResNet-18 to capture discriminative features of the image thus, help identifying falsified/irrelevant images. The modality fusion is conducted using late fusion approach to obtain an integrated representation that boost the performance.

In the experimental verification part, it can be clearly observed from the results that the proposed system attains superior performance, as it achieves a test accuracy and F1-score of 98.63%, reflecting high levels of precision and recall in both fake and real news classes. The decision of using a balanced dataset and standard performance evaluation parameters (accuracy, precision, recall, F1-score) are made for fair judgment of the proposed system. In addition, a correlation between the results of training and validation signifies that the proposed system has good generalization capability.

One important aspect of this paper is the development of an AI based validation layer, which add a dimension of reasoning into the detection. As opposed to standard ML based models, where only learned representations are taken into consideration, this layer analyzes semantics, analyzes linguistic features and analyzes the credibility of the sources as an independent check of the produced prediction. The development of this layer converts the system into a hybrid where data based predictions are augmented by logic-based validation, and the predictions are both more accurate and more interpretable.

Introducing a three-layered decision pipeline adds another layer of robustness. Here the decision engine is essentially using the results from both the ML model as well as the AI validation layer to double check the output of the predictions, prior to final decision making. In situations where the two outputs match a greater level of confidence can be assigned to the prediction. Conversely, if there is a conflict between the two, decision is made based on the confidences of both models and the strength of reasoning.

Usability and practical applicability were a key design consideration for the system. Multilingual input, confidence display, and a clean user interface created using Streamlit provide for effective interaction and can be seamlessly integrated in a real-world system. These functionalities make the system adaptable for multiple uses such as news verification sites, social media analysis and filtering, digital journalism, and content management/filtering systems. The ease-of-use coupled with detailed explanations allows the user to build trust in the system regardless of technical aptitude.

As well as the above features, this proposed system is a solid starting point for further developments. Future work on this system could look at the integration of more sophisticated multimodal fusion approaches (cross-modal, attention-based etc.), real-time data streams, and the addition of explainable AI to make the system even more understandable. Finally, to increase its impact at a broader level, it is conceivable that the system could be expanded for large scale deployment or integrated with automated fact checking pipelines.

In summary, our proposed Multimodal Fake News Detection system overcomes the shortcomings of traditional unimodal as well as purely machine learning-based solutions, yielding a more reliable, interpretable, and scalable system. By combining deep learning on multiple modalities with an AI based reasoning process, it stands as a viable solution to tackle modern day misinformation with a greater degree of certainty.

8.2 FURTHER ENHANCEMENTS

The Multimodal Fake News Detection system presented in this work has shown high performance, reliability and generalization power. However, there is room for further improvement and extension. In the face of increasing diversity and volume of misinformation, the future direction of the model is directed towards increasing its intelligence, scalability, interpretability and robustness to adapt to diverse environments and application needs.

The next direction for further development is based on advanced image processing approaches. In the proposed model, Convolutional Neural Networks are used. However, the more advanced models, like Vision Transformer (ViT), would be better in terms of visual recognition because ViT can learn long-range dependencies between regions in an image. Besides that, the cross-modal attention could be employed to further investigate interactions between textual features and visual features for robust detection of complex inconsistency and manipulation.

A further important feature would be the addition of real-time fact checking facilities. Connections with online fact checking APIs and structured knowledge bases would allow active rather than pattern-based fact checking to take place. Such connections would allow the assertions to be cross referenced with verified data in real-time and significantly improve the trustworthiness of the system in critical contexts, such as for the press and public information systems.

Regarding the deployment it is also of great importance to optimize the model and make it suitable for deployment in a real-world scenario. In a further step, we could deploy the system on a cloud environment using GPU to compute more efficiently on very large amount of data and achieve real-time inference at a low latency. We can investigate on model compression and quantization techniques to speed up the model and work on distributed processing.

Model explainability is also a key area of concern. By applying Explainable AI (XAI) methods (i.e. Attention visualization, feature attribution, and saliency mapping) a greater insight could be developed into the underlying model's decision making process which will improve transparency and user confidence and is required for any sensitive application.

Exploring sophisticated multimodal fusion schemes also provides a potential research pathway. The existing system utilizes a late fusion method, while a transition to an attention based fusion scheme, such as cross-attention and co-attention models, may provide deeper interactions between modalities. This may lead to a better modeling of the relationships between text and image, which is crucial to dealing with misdirections relying on cross-modal mismatches.

In order to increase applicability to the real world even further, the system could be connected to social networks to constantly track and analyze the contents in social media. If integrated with social networks, the system would be able to detect fake news in real-time when it is being distributed, and would hence prove immensely useful for combating and tracking misinformation.

Accessibility wise browser extensions and mobile apps can also make system accessible for broader audience. An end user could verify the news content directly from the browser or mobile phone.

Thirdly, continuous learning framework has been built up. Through applying the techniques of online learning or adaptive updating models, it is enabled to adapt to the emerging new trends of misinformation and the model keeps up with the speed of the development of fake news trends.

In summary, the above improvements will lead to a more scalable, intelligent, real-time fake news detection system. With the development of sophisticated multimodal learning techniques, real-time verification methods and user-oriented features, the system will become a holistic solution to tackle the problems of misinformation detection and analysis and enhance the overall trustworthiness and credibility of information.

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APPENDIX A: SOURCE CODE SNIPPETS

1. PREDICTION CODE SNIPPET

```
//Prediction Code
import os
import sys
import torch

sys.path.insert(0, os.path.dirname(os.path.abspath(__file__)))

from models.multimodal_model import MultimodalFakeNewsDetector, get_tokenizer
from utils.image_utils import download_image, pil_to_tensor, get_placeholder_tensor

MODEL_PATH = os.path.join(os.path.dirname(__file__), "models", "saved", "best_model.pt")
DEVICE     = torch.device("cuda" if torch.cuda.is_available() else "cpu")

_model = None


def load_model():
    global _model
    if _model is not None:
        return _model
    if not os.path.exists(MODEL_PATH):
        return None
    model = MultimodalFakeNewsDetector(dropout=0.0).to(DEVICE)
    ckpt  = torch.load(MODEL_PATH, map_location=DEVICE)
    model.load_state_dict(ckpt["model_state"])
    model.eval()
    _model = model
    return _model


@torch.no_grad()
def predict(title: str, image_url: str = ""):
    """
    Returns dict with:
      - label       : "FAKE" or "REAL"
      - confidence  : float 0-1
      - fake_prob   : float 0-1
      - real_prob   : float 0-1
      - model_loaded: bool
    """
    model = load_model()
```

```
if model is None:
    return {
        "label":        "UNKNOWN",
        "confidence":   0.0,
        "fake_prob":    0.5,
        "real_prob":    0.5,
        "model_loaded": False
    }

tokenizer = get_tokenizer()
enc = tokenizer(
    [title],
    padding=True,
    truncation=True,
    max_length=128,
    return_tensors="pt"
)
input_ids      = enc["input_ids"].to(DEVICE)
attention_mask = enc["attention_mask"].to(DEVICE)

# Image
img          = download_image(image_url) if image_url else None
image_tensor = pil_to_tensor(img).unsqueeze(0) if img else
get_placeholder_tensor().unsqueeze(0)
image_tensor = image_tensor.to(DEVICE)

logits = model(input_ids, attention_mask, image_tensor)
probs  = torch.softmax(logits, dim=1)[0].cpu().numpy()

fake_prob  = float(probs[0])
real_prob  = float(probs[1])
label      = "REAL" if real_prob > fake_prob else "FAKE"
confidence = max(fake_prob, real_prob)

return {
    "label":        label,
    "confidence":   confidence,
    "fake_prob":    fake_prob,
    "real_prob":    real_prob,
    "model_loaded": True
}
```

APPENDIX B: PUBLICATION

The research outcomes of this project have been documented as an IEEE-format paper titled:

"Multimodal Fake News Detection Using AI-Driven Deep Learning With CNN And Transformer-Based Fusion"

Author: Sridev S, School of Computer Science and Engineering, Vellore Institute of Technology, Vellore, India.

Guide: *Prof.* Vijayasherly V, School of Computer Science and Engineering, Vellore Institute of Technology, Vellore, India.

The paper presents the complete system design, methodology, implementation, and experimental evaluation of the proposed multimodal fake news detection system. Key findings indicate that the model achieves high performance with an accuracy of 98.3%, F1-score of 98.3%, precision of 99.0%, and recall of 98.6%, demonstrating the effectiveness of the multimodal approach combined with AI-based validation for reliable and accurate fake news detection.